

Your Cancer Summary Report

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Stage IV High-Grade Serous Ovarian Cancer with BRCA1 Germline Mutation

Precision Oncology Research Center

Report Date: March 27, 2026

⚠️ DRAFT - FOR CLINICIAN REVIEW

This report has not yet been reviewed by your healthcare team. Please wait for your doctor to discuss these results with you.

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At a Glance

Your Diagnosis

Ovarian Cancer - High-Grade Serous Carcinoma (HGSOC)

Stage: Stage IV | **Grade:** High Grade

You have an advanced form of ovarian cancer (Stage IV) that has spread beyond the ovaries. While this is serious, your tumor has specific genetic features — including an inherited BRCA1 mutation and a high genomic instability score (HRD 54) — that make it particularly sensitive to PARP inhibitor therapy, a targeted treatment with proven survival benefit in your situation.

✓ Positive Factors

- BRCA1 germline mutation qualifies you for PARP inhibitor maintenance therapy (olaparib — FDA Approved OlympiA/SOLO-1 trials)
- High HRD score (54, above the 42 threshold) confirms your tumor is genomically unstable and sensitive to DNA-damaging treatments
- VEGFA is spatially active — you are already receiving bevacizumab targeting this pathway
- PIK3CA E545K and PTEN loss create a PI3K/AKT vulnerability that may be targetable
- Substantial CD8 T cell infiltration (82 spots, 9%) suggests partial immune activation

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- RB1 and CDKN2A deletions suggest aggressive cell cycle dysregulation
- Stage IV disease requires sustained systemic therapy
- High CA-125 (389 U/mL) reflects active disease burden

What Your Tests Found

Genetic testing of your tumor revealed the following important findings:

BRCA1

Pathogenic

Germline pathogenic mutation (VCF-confirmed HGSOC context)

You inherited a change in the BRCA1 gene that disrupts the cell's ability to repair DNA breaks. This same weakness that increases cancer risk also makes the cancer highly vulnerable to PARP inhibitor drugs that block an alternative repair pathway — a principle called synthetic lethality.

Treatment Option: Olaparib 300mg BID maintenance — FDA Approved (SOLO-1 trial: 60% reduction in disease progression risk)

TP53

Pathogenic

R175H missense (AF 73%, COSV57271936)

The TP53 gene — the genome's main guardian — carries a hotspot mutation found in 96% of HGSOC tumors. This confirms the tumor's HGSOC identity and explains its aggressive behavior. APR-246 (eprenetapopt), a drug that restores mutant p53 function, is in Phase III trials.

Treatment Option: APR-246/eprenetapopt (Phase III, investigational) — targets mutant p53 R175H

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PIK3CA

Pathogenic

E545K missense (AF 42%, COSM55514880)

A PIK3CA mutation was found that activates a cell growth pathway. Combined with PTEN loss (see below), this creates a double activation of the PI3K/AKT/mTOR growth signaling chain. Alpelisib (FDA-approved in breast cancer) and copanlisib are being studied in HGSOc.

Treatment Option: Alpelisib off-label or clinical trial; copanlisib (Phase II in ovarian cancer)

PTEN

Pathogenic

LOH splice acceptor (AF 85%) + genomic deletion ($\log_2 = -1.45$)

Both copies of the PTEN gene have been lost. PTEN normally puts the brakes on the PI3K/AKT growth pathway. Without it, combined with the PIK3CA mutation, this pathway is hyperactivated. AKT inhibitors (capivasertib) are in Phase III trials for this combination.

Treatment Option: Capivasertib AKT inhibitor (Phase III CAPItello-291)

CCNE1

Clinically Significant — Platinum Resistance Marker

Amplification, CN=6, $\log_2 = 1.58$ (chr19q12)

Extra copies of the CCNE1 gene (cyclin E1) are present in your tumor. This is a known marker that can predict reduced sensitivity to platinum-based chemotherapy (carboplatin). Your care team should monitor your response to the current regimen closely. CDK2 and Wee1 inhibitors specifically target CCNE1-amplified tumors.

Treatment Option: Wee1 inhibitor (adavosertib, Phase II); CDK2 inhibitor (investigational)

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MYC

Clinically Significant

Amplification, CN=5, log2=1.32 (chr8q24)

Extra copies of the MYC gene drive rapid tumor growth. This is found in 30% of HGSOC. BET inhibitors targeting MYC are in preclinical and early clinical development.

Treatment Option: BET inhibitors (preclinical/early Phase I)

RB1/CDKN2A

Clinically Significant

Homozygous deletions (RB1 log2=-1.85, CN=0; CDKN2A log2=-2.12, CN=0)

Both copies of two important cell-cycle checkpoint genes have been deleted. This completely removes the brakes on cell division, driving aggressive tumor growth. This finding also raises caution about CDK4/6 inhibitor use, as RB1 loss may cause resistance.

Treatment Option: Note: RB1 loss may predict CDK4/6 inhibitor resistance — avoid palbociclib

HRD Score

HRD-positive (score 54 $\geq$ 42) — PARP eligible by both BRCA and HRD-score routes

Score 54 (LOH=18, TAI=15, LST=21) — HRD-positive

A genomic scar test measuring DNA repair damage (HRD score 54) came back positive — above the threshold of 42. Combined with your BRCA1 mutation, you qualify for PARP inhibitor therapy through two independent pathways. This is a strong indicator that olaparib or niraparib maintenance would be effective.

Treatment Option: PARP inhibitor maintenance confirmed via both BRCA1 mutation AND HRD score (olaparib, niraparib)

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Your Tumor Environment

Advanced spatial analysis shows how different cells are organized in your tumor:

Tumor Microenvironment

900-spot Visium spatial transcriptomics across HGSOC tumor. Cell type composition: 181 tumor cells (20%), 164 fibroblasts (18%), 151 endothelial (17%), 145 mesenchymal (16%), 82 CD8 T cells (9%), 80 macrophages (9%), 70 regulatory T cells (8%), 27 CD4 T cells (3%). Mixed immune microenvironment. VEGFA spatially active (Moran's $I=0.092$). High mesenchymal/fibroblast component suggests active desmoplastic stroma.

Immune Status: • Mixed immune activity - Variable immune response across the tumor

A detailed map of your tumor shows that it has a rich mix of cell types, including immune cells that are trying to fight the cancer and structural cells that are somewhat blocking them. The drug you're already on (bevacizumab) targets a growth factor (VEGFA) that our analysis shows is actively being produced across the tumor. There are immune cells present, but they are partially held back by the tumor's surrounding tissue — this is being studied as a target for future immunotherapy combinations.

Key Patterns Observed

- VEGFA spatially patterned (Moran's $I=0.092$) — supports bevacizumab targeting of active angiogenic regions
- MYC weakly patterned across tumor (Moran's $I=0.080$) — proliferative zones distributed throughout
- CD8 T cells present (82 spots, 9%) but partially balanced by Tregs (70 spots, 8%) — CD8:Treg ratio 1.17 indicates immune suppression
- High fibroblast/mesenchymal fraction (22% combined) suggests immune

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- TP53 expression weakly patterned — consistent with near-universal TP53 mutation in HGSOC

Treatment Options

Based on your test results, your care team may consider these treatments:

PARP Inhibitor Maintenance — Olaparib

targeted

FDA Approved

Expected Response: 60% reduction in disease progression (SOLO-1 HR=0.30)

After completing platinum-based chemotherapy, olaparib is taken twice daily as maintenance therapy. It exploits the BRCA1 mutation to kill residual cancer cells that cannot repair DNA damage. In the SOLO-1 trial, olaparib maintenance extended the time before cancer progression by nearly 3 years in BRCA-mutated ovarian cancer patients.

Why this may help you: BRCA1 germline mutation plus HRD score 54 (positive). Dual-route eligibility (BRCA-mutation route AND HRD-score route). SOLO-1 trial demonstrated unprecedented progression-free survival benefit in newly diagnosed BRCA-mutated HGSOc.

Common side effects: Nausea (manage with anti-emetics), Fatigue, Anemia, Low platelet count, Rarely: myelodysplastic syndrome (long-term monitoring required)

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Bevacizumab (Anti-VEGF) — Continue

targeted

FDA Approved

Expected Response: 4-month PFS improvement (GOG-0218/ICON7 trials)

Bevacizumab is already in your regimen. It blocks blood vessel growth that feeds the tumor by targeting the VEGFA protein. Your spatial transcriptomics data shows VEGFA is actively and heterogeneously expressed in your tumor, supporting continued anti-VEGF therapy.

Why this may help you: VEGFA spatially expressed and weakly clustered across tumor microenvironment (Moran's I=0.092). Current treatment appears biologically rational. Continuation with PARP inhibitor combination (bevacizumab + olaparib, PAOLA-1 trial) shows additional benefit.

Common side effects: High blood pressure, Wound healing complications, Rare: bowel perforation or blood clots, Fatigue, Protein in urine

PI3K/AKT Pathway Inhibition — Clinical Trial

targeted

Clinical Trial Only

Expected Response: ~30-40% in PI3K-altered HGSOC (early data)

Because your tumor has both a PIK3CA mutation and complete PTEN loss, the PI3K/AKT pathway is doubly activated. Capivasertib (an AKT inhibitor in Phase III) or alpelisib (PI3K inhibitor, FDA-approved in breast cancer) may be options either on trial or off-label.

Why this may help you: PIK3CA E545K pathogenic mutation (AF=42%) plus complete PTEN loss (both alleles deleted). Stouffer meta-analysis $q \approx 0$ confirms convergent pathway activation signal across modalities. AKT2 amplification further reinforces PI3K/AKT dependence.

Common side effects: High blood sugar (especially alpelisib), Diarrhea

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Clinical Trials

You may be eligible for these research studies offering new treatments:

SOLO-1 Maintenance Extension: Olaparib in Newly Diagnosed BRCA-Mutated Advanced Ovarian Cancer

Phase: 3 | **Status:** Active, follow-up

Trial ID: NCT03402841

This is the landmark trial that demonstrated olaparib maintenance extends survival in newly diagnosed BRCA-mutated HGSOc — exactly your situation. The trial team is monitoring long-term outcomes and follow-up cohorts may be enrolling.

CAPitello-291: Capivasertib + Paclitaxel in PIK3CA/PTEN/AKT-Altered Solid Tumors

Phase: 3 | **Status:** Active, enrolling

Trial ID: NCT03855254

You have PIK3CA E545K mutation AND complete PTEN loss — the precise molecular combination targeted by capivasertib in this trial. This is a strong candidate for enrollment discussion with your oncologist.

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Your Monitoring Plan

Regular check-ups help us track your health and catch any changes early.

Scheduled Tests

Every 3 months	CA-125 tumor marker Primary marker for treatment response and relapse detection in HGSOC; baseline CA-125 is elevated at 389 U/mL
Every 6 months or when CA-125 rising	CT chest/abdomen/pelvis Radiographic assessment of disease extent and treatment response
Monthly during chemotherapy + PARP inhibitor maintenance	CBC with differential Monitor for anemia, thrombocytopenia, and neutropenia — common with olaparib + carboplatin
Before each chemotherapy cycle	Renal function (creatinine, eGFR) Carboplatin dosing is kidney-function dependent; bevacizumab can affect renal function
Every clinic visit while on bevacizumab	Blood pressure Bevacizumab commonly causes hypertension requiring management
Every 6 months	ctDNA liquid biopsy (BRCA1 reversion, PIK3CA E545K) Early detection of acquired resistance mutations (BRCA1 reversion mutations predict PARP inhibitor resistance)

⚠ When to Call Your Doctor

Contact your care team right away if you experience:

- Sudden severe abdominal pain or distension (possible bowel perforation on bevacizumab — call 911)

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- Fever above 38°C/100.4°F (possible neutropenic fever — go to ER immediately)
- Persistent nausea or inability to eat for more than 2 days
- New or worsening leg swelling (possible DVT)
- Decreased urination or severe fatigue (possible kidney problem)

Questions? Gynecologic oncology nurse navigator — available 24/7 for urgent concerns

For Your Family

Your BRCA1 mutation is inherited, meaning your biological children, siblings, and parents may carry the same mutation. BRCA1 mutation carriers have significantly elevated lifetime risk of ovarian cancer (~44%) and breast cancer (~72%). Family members who test positive can undergo enhanced screening or risk-reducing surgeries before cancer develops.

Recommended Steps for Family Members

- Schedule appointment with a certified genetic counselor within 4 weeks
- Inform daughters, sons, sisters, and brothers about the BRCA1 finding and recommend genetic testing
- For BRCA1-positive daughters: discuss risk-reducing bilateral salpingo-oophorectomy after age 35-40 and enhanced breast MRI screening
- Contact FORCE (facingourrisk.org) for peer support from others with BRCA1 mutations

Genetic Counseling Recommended: A genetic counselor can help explain what these results mean for your family members and discuss testing options.

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Resources & Support

You don't have to face this alone. These resources can help:



FORCE (Facing Our Risk of Cancer Empowered)

Premier nonprofit for hereditary breast and ovarian cancer (HBOC) community. Provides peer support, educational webinars, clinical trial matching, and family planning resources specific to BRCA mutation carriers.

📞 1-866-288-7475 | 🌐 <https://www.facingourrisk.org>



Ovarian Cancer Research Alliance (OCRA)

Leading ovarian cancer advocacy organization with patient support programs, clinical trial navigator, and community for HGSOC patients and caregivers.

📞 1-212-268-1002 | 🌐 <https://ocrahope.org>



Cancer Financial Assistance Coalition (CFAC)

Connects patients to copay assistance programs for olaparib (AstraZeneca Patient Assistance), bevacizumab, and clinical trial participation support including travel assistance.

| 🌐 <https://www.cancerfac.org>

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Data Sources: mockepic-clinical, fgbio-sequencing, mcp-genomic-results, mcp-spatialtools, mcp-multiomics-stouffer, mcp-patient-report

Report Version: 1.0

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